

Solution Requirements

Date	30 September 2026
Team ID	SWTID-2026-8336
Project Name	EduGenie — Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

Define the requirements of **EduGenie** to clearly explain what the system should do and how well it should work. Functional requirements describe the main features of EduGenie, such as answering student doubts, explaining topics, creating summaries and quizzes, and tracking learning progress. Non-functional requirements describe the quality of the system, such as being fast, easy to use, secure, reliable, and responsive.

Functional Requirements (FR): Define the specific functions and features that EduGenie must provide to help students learn, clear doubts, practice, and track their progress.

Non-Functional Requirements (NFR): Define the quality standards that EduGenie should maintain, including performance, security, usability, reliability, scalability

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Students can securely register and log in using email and password. Passwords are protected using bcrypt and authenticated sessions use JWT.</i>	High
2	Authorization levels	<i>The system provides authenticated student access to learning features. Only logged-in users can access protected learning, profile, settings, quiz results, and progress</i>	High

		<i>features.</i>	
3	External interfaces	<i>The system connects the React.js frontend with the Node.js and Express.js REST API and integrates with the Google Gemini API for AI-powered learning support and MongoDB Atlas for data storage.</i>	High
4	Transactions processing	<i>The system processes user authentication, learning requests, AI requests, quiz generation, answer evaluation, learning-history updates, study-plan data, and progress data through the backend and database.</i>	High
5	Reporting	<i>The system provides quiz results, answer evaluation, learning history, study-plan information, and study progress to help students review their learning.</i>	Medium
6	Business rules, etc.	<i>Students must be authenticated to use protected learning features. AI-generated learning support is provided through Google Gemini. Quiz answers are evaluated and results are stored for progress tracking.</i>	High
7	Compliance to laws or regulations	<i>The application should protect user and learning data and follow applicable institutional privacy and data-protection requirements.</i>	Medeium

8	Other	<i>The system provides learning history, profile management, settings, forgot-password support, quiz results, and answer evaluation.</i>	Medium
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The system should provide responsive page navigation and API interactions under normal usage conditions.</i>	<i>Responsive operation under normal local usage. No formal performance benchmark has been conducted.</i>
2	Scalability	<i>The system should support increasing numbers of students and learning data through scalable backend and MongoDB Atlas infrastructure.</i>	<i>Scalability under high concurrent users has not been formally tested.</i>
3	Security & Data Privacy	<i>User authentication must use JWT, passwords must be protected using bcrypt, and user and learning data stored in MongoDB Atlas should be protected from unauthorized access.</i>	<i>Email/password authentication works; JWT and bcrypt are implemented; protected features require authentication.</i>
4	Reliability & Availability	<i>The application should reliably perform authentication, AI learning requests, quiz operations, and learning-data operations during normal use.</i>	<i>Functional browser/manual testing completed. No formal uptime or fault-tolerance testing conducted.</i>

5	Usability & Accessibility	<i>The interface should be simple, clear, and easy for students to use for asking doubts, learning topics, generating quizzes, viewing results, and tracking progress.</i>	<i>Main learning functions should be accessible through clear navigation and understandable user-interface elements.</i>
6	<i>Other</i>	<i>The system should be maintainable and modular, with separate frontend, backend, database, authentication, and AI integration components.</i>	Features should remain organized so that future learning features and AI capabilities can be added without major restructuring.